USN | | | | | | | | | | [23BSMA12]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

COMPUTATIONAL MATHEMATICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ii. Answer all the questions

		PART-A	**05 X 02 =10**		
1.	**a.**	If $A = \{2,4,6\}$, then find the cardinality of the power set A?	L-1	CO1	**02**
	b.	Determine the nature of the proposition: p ∧ ~p.	L-2	CO1	**02**
	c.	Find the first order partial derivatives of $u = e^x \sin y$.	L-2	CO1	**02**
	d.	Determine Eigenvalues of the matrix = $\begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.	L-1	CO1	**02**
	e.	Solve the solution of $(D^2 - 1)y = 0$.	L-3	CO5	**02**
		PART-B	**06 X 05 = 30**		
2.	**a.**	Explain Injective and Surjective function with suitable examples.	L-2	CO1	**05**
		OR			
	b.	If $A = \{2,4,6,8\}$ and $B = \{1,2,3\}$ and relations R_1, R_2, R_3, R_4 from A to B be defined as follows: **i.** aR_1b if $a \leq b$ **ii.** aR_2b if $a > b$ **iii.** aR_3b if a divides b **iv.** aR_4b if b divides a Then find the relations.	L-3	CO1	**05**
3.	**a.**	Show that $p \to q$ and $\neg p \vee q$ are logically equivalent.	L-3	CO3	**05**
		OR			
	b.	Determine the truth value of each of the following quantified statements, the universe being the set of all non-zero integers. **i.** $\exists x, \exists y, [xy = 1]$ **ii.** $\exists x, \forall y, [xy = 1]$ **iii.** $\forall x, \exists y, [xy = 1]$ **iv.** $\exists x, \exists y, [(2x + y = 5) \wedge (x - 3y = -8)]$ **v.** $\exists x, \exists y, [(3x - y = 17) \wedge (2x + 4y = 3)]$	L-3	CO3	**05**
4.	**a.**	If $u = (x-y)^4 + (y-z)^4 + (z-x)^4$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	L-3	CO6	**05**

		OR			
	b.	Find the directional derivative of the function $f = x^2yz + 4xz^2$ at $(1,-2,-1)$ along the direction of the vector $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$.	L-3	CO6	**05**
5.	**a.**	Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ by converting it into echelon form.	L-3	CO1	**05**
		OR			
	b.	Test for consistency $$5x_1 + x_2 + 3x_3 = 20$$ $$2x_1 + 5x_2 + 2x_3 = 18$$ $$3x_1 + 2x_2 + x_3 = 14$$	L-3	CO1	**05**
6.	**a.**	Solve $(D^2 - 4D + 4)\,y = e^{2x}$.	L-3	CO5	**05**
		OR			
	b.	Find the solution of $(D^2 - 2D + 2)y = \cos(x-1)$.	L-3	CO5	**05**
7.	**a.**	Investigate the extreme values of the given function $f(x,y) = {}^3 + y^3 - 3axy$.	L-3	CO6	**05**
		OR			
	b.	Write the Python code to find the first and second order partial derivatives of the function $f(x,y) = x^3 - 2x^2y^3$.	L-3	CO6	**05**
		PART-C	**06 X 10 = 60**		
8.	**a.**	Show that the following relation R defined on the set of all integers Z is an equivalence relation. $R = \{(x,y):\ x,y \in Z \text{ and } (x-y) \text{ is an even number}\}$.	L-3	CO1	**10**
		OR			
	b.	Write Python program to create the following sets A = {1,2,3,4,5,6} and B = {Hari, Prem, 102.55, 23, 12}. And also write python code for the following **i.** cardinality of A and B **ii.** add 21, 22 to the set A **iii.** Remove 23, Hari from set B	L-3	CO2	**10**
9.	**a.**	Prove the following logical equivalence without using Truth table. **i.** $p \vee [p \wedge (p \vee q)] \Leftrightarrow p$ **ii.** $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \Leftrightarrow (p \vee q \vee r)$	L-3	CO3	**10**

		OR			
	b.	Write a python code for negation, conjunction, disjunction and conditional of two propositions p and q.	L-3	CO3	**10**
10.	**a.**	Find $div\,\vec{v}$ and curl $\vec{v}$ where $\vec{v} = \nabla(x^3 + y^3 + z^3 - 3xyz)$.	L-3	CO6	**10**
		OR			
	b.	Prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$, where $v = \frac{1}{(x^2+y^2+z^2)^{\frac{1}{2}}}$.	L-3	CO6	**10**
11.	**a.**	Find all the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.	L-3	CO4	**10**
		OR			
	b.	Write a python code to find all the eigen values and corresponding eigen vectors of $3X3$ matrix. Also write the condition so that two matrices are similar?	L-3	CO2	**10**
12.	**a.**	Solve $(D^3 - D^2 - 4D + 4)y = 0, y(0) = 1, y'(0) = 2\ and\ y''(0) = 1.$	L-3	CO5	**10**
		OR			
	b.	Solve $(D^2 + 4)y = 4\,sin2x + cos5x.$	L-3	CO5	**10**
13.	**a.**	Write python code to solve the following differential equation $(D^3 - D^2 - 4D + 4)y = 0,\ y(0) = 1,\ y'(0) = 2$ and $y''(0) = 1.$	L-3	CO4	**10**
		OR			
	b.	Solve the differential equation $(D^2 + D)\,y = x^2 - 2x - 16.$	L-3	CO5	**10**